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Early lexico-semantic modulation of motor related areas during action and non-action verb processing

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ABSTRACT

Although action verb processing deficits have been described in diseases affecting the motor system, research on temporal processing in this area has not been reported. In this study, action and non-action verb processing was contrasted in healthy volunteers using electro-encephalography. These data may serve as a control condition for further research in motor disorders. Latency and amplitude evaluations as well as source reconstruction were applied on event-related potentials. Action verbs evoked higher activation in bilateral sensorimotor areas from 155 to 174 ms and in bilateral dorsolateral prefrontal cortex (DLPFC) from 219 to 238 ms. Hand action verb processing activates the motor programmes of the actions the verbs refer to. This seems not restricted

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to the core (pre)motor cortical areas of the brain. A broad motor brain network is hypothesized to be involved. While sensorimotor activation seems essential for action verb understanding, this cannot be concluded for DLPFC activation.

1. Introduction

Substantial research has been conducted on the perception of action related linguistic material such as action verbs. Besides the classic language areas, also the premotor and primary motor cortex are reported to be involved in the processing of action-related words and sentences. Moreover, this processing appears to occur in a somatotopic way. Action verbs related to face, arm or leg movements elicit the strongest activation close to the cortical motor representation of the face, hands or legs respectively (Aziz-Zadeh, Wilson, Rizzolatti, & Iacoboni, 2006; Boulenger, Hauk, & Pulvermüller, 2009; Buccino et al., 2005; Hauk, Johnsrude, & Pulvermüller, 2004; Kemmerer, Castillo, Talavage, Patterson, & Wiley, 2008; Pulvermüller, Härle, & Hummel, 2001; Pulvermüller, Shtyrov, & Imoniemi, 2005; Raposo, Moss, Stamatakis, & Tyler, 2009; Repetto, Colombo, Cipresso, & Giuseppe, 2013; Shtyrov, Hauk, & Pulvermüller, 2004; Tettamanti et al., 2005). Although this somatotopical activation is not always found (Arévalo, Baldo, & Dronkers, 2012; Postle, Ashton, McFarland, & de Zubicaray, 2013), a review by Kemmerer and Gonzalez-Castillo (2010) showed surprising consistencies among different labs and languages.

Unfortunately, the underlying mechanism responsible for the motor activation remains a contentious issue because conflicting results are found on the processing stage during which this motor activation occurs. Some studies revealed somatotopic motor activation after auditory and visual single word presentation from 130 to 170 ms (Pulvermüller et al., 2005; Shtyrov et al., 2004) and 210 to 230 ms (Hauk & Pulvermüller, 2004b) respectively. In addition, visually presented action words appear to interfere with a reaching movement already within 200 ms after word onset (Boulenger et al., 2006). Within the first 200 to 250 ms after word presentation, essential lexical and semantic processes are known to occur (Federmeier & Kutas, 2001; Hauk, Coutout, Holden, & Chen, 2012; Penolazzi, Hauk, & Pulvermüller, 2007). Thus, actions and action semantics related to words apparently share cognitive and neural resources. This is in line with theories of embodied cognition which state that all concepts are (partly) modality dependent and are grounded in neural action and perception systems (e.g. Barsalou, 1999; Dove, 2009). Consequently, motor areas are suggested to be involved in lexical access (Hauk, Shtyrov, & Pulvermüller, 2008). By contrast, other studies found a much later motor cortex modulation around 500 ms post-stimulus onset (Oliveri et al., 2004; Papeo, Vallesi, Isaja, & Rumiati, 2009). At this stage, post conceptual processes of word recognition occur (Marinkovic et al., 2003). Motor strip activation would then follow the identification of the action concept, instead of being part of it. This 'spreading activation' occurs because the word's concept is associated with the motor system controlling the respective action (Hickok, 2010) or because of mental imagery (Tomasino, Fink, Sparing, Dafotakis, & Weiss, 2008).

A recent study conducted by Moseley, Pulvermüller, and Shtyrov (2013) used excellent equipment to elucidate which processing stage is involved. Passive reading of written words was found to evoke maximal brain responses at 150 ms post-stimulus onset. Besides widespread activity in perisylvian regions for all words, inferior frontal gyrus and precentral cortex were significantly more engaged during action compared to abstract word processing. Thus, category-specific semantics seem to be represented in the neural systems for perception and action. As these regions were activated within the first 200 ms, this representation seems essential for concept understanding. Unfortunately, while the action words were mostly verbs, the abstract words were a conglomeration of both nouns and verbs. Although grammatical class in itself does not have an influence on the organization of knowledge in the brain (Vigliocco, Vinson, Druks, Barber, & Cappa, 2011), neurophysiological differences between verbs and nouns have been reported (Kellenbach, Wijers, Hovius, Mulder, & Mulder, 2002; Osterhout, Bersick, & McKinnon, 1997). Thus, a possible lexical/grammatical confound cannot be excluded to have influenced the results.

In sum, there is no consensus on the function, timing and necessity of motor cortex activation during action related word processing. Diseases affecting the motor system might help in clarifying this issue. If the motor cortex contributes to word understanding, action verb processing deficits should occur in patients with disturbances of their motor system. Indeed, a large variety of pathologies have been shown to evoke disturbances in action verb processing: motor neuron disease (Bak & Hodges, 1999, 2004; Bak, O'Donovan, Xuereb, Boniface, & Hodges, 2001; Grossman et al., 2008), progressive supranuclear palsy (Bak et al., 2006; Daniele, Giustolisi, Silveri, Colosimo, & Gianotti, 1994), fronto-temporal dementia (Cappa et al., 1998), aphasia (Saygin, Wilson, Dronkers, & Bates, 2004), apraxia in chronic stroke patients (Buxbaum & Saffran, 2002), lesions in the right frontal area (Neininger & Pulvermüller, 2003), and Parkinson's disease (PD) (Fernandino et al., 2012). Unfortunately, most of these studies contrasted action verbs with non-action *nouns.* As mentioned above in relation to the Moseley et al. (2013) study, neurophysiological differences between verbs and nouns have been reported (Kellenbach et al., 2002; Osterhout et al., 1997). In addition, verbs are inherently more difficult than nouns because of more complex semantic and syntactic constraints (for a review, Druks, 2002). Therefore, these action verb processing deficits might rather be related to grammatical than to semantic aspects. Moreover, no temporal information on action linguistic processing in motor pathologies is available. To our knowledge, all studies reported behavioural and neuroimaging data with good spatial, but poor temporal resolution like e.g. fMRI. However, by applying neurophysiological tools such as electro-encephalography (EEG), one could elucidate which processing stage is affected in these motor pathologies and consequently, which processing stage relies (partly) on motor related brain areas.

Therefore, the present study aimed at evaluating motor related brain activations during action verb processing in motor pathologies by use of EEG. All action verbs denoted movements performed with hand and/or arms to evoke focalized activity in motor cortex. To overcome a grammatical class confound, these action verbs should be contrasted with another group of verbs. As contrast condition, non-action verbs were chosen instead of action verbs related to another body part because these verbs require no or only limited motor involvement. Variability in disease severity will cause variability in motor cortex deficiency. If the control condition would rely on motor cortex activity, variability in its processing would occur as well. A control condition should however provide a reliable comparison for the measure of interest. If the control condition varies, no straightforward conclusion can be made about the measure of interest.

To our knowledge, no EEG research has been performed in which action verbs were contrasted with a group of only non-action verbs, not even in healthy populations. Therefore, the task was first administered in a group of healthy control participants. These data are presented in the present study. They will be used as a control condition for further experiments in patient populations with motor disorders. Therefore, an accessible method to use in a clinical setting was developed.

2. Methods

2.1. Participants

30 (male/female: 22/8) healthy, right-handed (Oldfield, 1971) volunteers (mean age ± standard deviation: 30.2 ± 10.6; age range: 18–57) were included in this study. They were all monolingual native speakers of Dutch and reported no history of hearing complaints, dyslexia or other speech-language problems, neurological or psychiatric disorders, and presented with normal or corrected-to-normal vision. None of them was on psycho-active drugs. All participants gave their written informed consent in accordance with the declaration of Helsinki. This study was approved by the local ethics committee.

2.2. Neurophysiological assessment

2.2.1. Stimuli

50 Action and 50 non-action verbs were selected from WordGen (Duyck, Desmet, Verbeke, & Brysbaert, 2004), based on the CELEX database (Baayen, Piepenbrock, & van Rijn, 1995). To evoke

focalized activity in sensorimotor cortices, all action verbs referred to hand and/or arm movements (e.g. to knead, to sew). The non-action verbs were abstract verbs unrelated to actions or body parts (e.g. to believe, to tolerate). A list of all stimuli items is provided in Appendix A. Both verb classes were as closely matched as possible on several psycholinguistic and lexical characteristics as to minimize their possible impact in early neurophysiological processing (Dambacher, Kliegl, Howmann, & Jacobs, 2006; Federmeier & Kutas, 2001; Hauk & Pulvermüller, 2004a; Hauk, Davis, Ford, Pulvermüller, & Marslen-Wilson, 2006; Hauk, Patterson, et al., 2006; Hauk et al., 2012; Penolazzi et al., 2007; Takashima, Ohta, Matsushima, & Toru, 2001). An overview of these features can be found in Table 1.

Semantic relatedness between verbs and body parts was determined in a pre-test by 11 native speakers of Dutch who did not participate in the EEG study. These body areas included (1) head (head/face/mouth), (2) arms (arms/hands/fingers), and (3) legs (legs/feet/toes). All verbs were scored in relation to these 3 body areas using a 5 point-scale ranging from 1, labelled "highly unrelated", to 5, labelled "highly related". Word imageability was estimated as well, following the same procedure. The question to be rated was: "how easily does this word evoke an image?" with 1 labelled "not at all" and 5 labelled "very easily". The only body part that was supposed to be associated with half of the verbs was 'arms'. This might have been noted by the participants and biased their scoring. Therefore, 30 leg and 30 head verbs were added in this pre-test. As these leg and head verbs only served as distractors, they were not specifically matched on psycholinguistic features to the verbs of the experimental set. They were randomly chosen from WordGen (Duyck et al., 2004) with as only requirement being related to legs/head respectively.

The arm action verbs were significantly more imaginable and more linked to arms than the non-action verbs. In addition, arm action verbs were more associated with arms than with legs and head. A similar finding was seen for the distractor verbs: leg and head verbs were significantly more linked

Table 1
Summary of stimulus characteristics. Mean ± SD is displayed. The *p*-value of the Mann–Whitney *U* test comparing action and non-action verbs is shown in the right column. For action verbs, the mutual comparison of the semantic relatedness scores for different body parts is shown on the left.

Feature		*Action verbs*	*Non-action verbs*	*P-value*
Word length				
- Letters		7.0 ± 1.3	6.9 ± 1.3	0.77
- Syllables		2.2 ± 0.4	2.3 ± 0.5	0.36
Word frequency		1.4 ± 0.6	1.6 ± 0.6	0.18
Bigramfrequency		12771 ± 3037	13811 ± 3129	0.06
Orthogr. neighborhood size		4.3 ± 4.0	4.5 ± 4.2	0.90
Imageability		4.5 ± 0.2	2.4 ± 0.6	***< 0.001***
Head relatedness	***< 0.001***[a]	1.5 ± 0.5	1.7 ± 0.8	
Arm relatedness		4.9 ± 0.1	1.4 ± 0.4	***< 0.001***[c]
Leg relatedness	***< 0.001***[b]	1.6 ± 0.5	1.2 ± 0.2	

[a] Arm action verbs are significantly more related to arms than to head

[b] Arm action verbs are significantly more related to arms than to legs

[c] Arm action verbs are significantly more related to arms than the non-action verbs

with legs and head respectively compared to other body parts and compared to the arm action and non-action verbs of the experimental set (Mann–Whitney *U* test: $p < 0.001$ for all comparisons).

2.2.2. Procedure

All arm action verbs and non-action verbs were presented in their infinitive form as single words to minimize the interference of syntactic processes. They were shown in black letters (font: Calibri; size: 96) on a white background in the middle of a computer screen that was placed 1 m in front of the participant. Stimuli were randomly presented with a stimulus frequency of 0.7 Hz (±1428 ms). No blank screen was shown in between successive stimuli. Participants were instructed to read each of the words mentally and to avoid overt articulation or any other kind of orofacial movement. To optimize EEG quality, they were encouraged to reduce eye-blinks as much as possible.

2.2.3. Data acquisition and analysis

EEG data were collected with Neuron-Spectrum-5 (4EPM) registration software (Neurosoft, Moscow, Russia). 21 Ag/AgCl electrodes were placed on the scalp according to the international 10/20 system (Fp1, Fpz, Fp2, F7, F3, Fz, F4, F8, C3, Cz, C4, T3, T5, T4, T6, P3, Pz, P4, O1, Oz, O2). Additional reference and ground electrodes were placed on the earlobes and forehead respectively. Neurophysiological data were recorded at a sampling rate of 500 Hz (0.05–75 Hz band-pass filter). Impedance of each electrode was kept below 5 kΩ.

Off-line EEG analysis was performed using BrainVision Analyzer 2 (Brain Products, Munich, Germany). After additional filtering (0.5–30 Hz band-pass filter, Notch filter 50 Hz), eye artefacts were excluded using Independent Component Analysis. Two components were removed (eye blinks; left-right eye movement) based on inspection of the components' spatial distribution (Joyce, Gorodnitsky, & Kutas, 2004; Mennes, Wouters, Vanrumste, Lagae, & Stiers, 2010). Next, the continuous EEG data were segmented into epochs of 1100 ms, starting 100 ms prior to stimulus onset, and baseline corrected to this pre-stimulus interval. Trials with voltage variations larger than 100 μV were manually rejected. By averaging over corresponding epochs, event-related potentials (ERPs) were computed for every subject, electrode, and verb category. ERP participant averages were then grand-averaged across participants for both verb classes separately.

2.2.4. ERP analysis

Visual presentation of single words typically evokes an ordered succession of 6 peaks. Whereas P1, N2 and P3 are known to be related to primary visual and visual attention processes (e.g. Di Russo, Martinez, Sereno, Pitzalis, & Hillyard, 2001; Folstein & Van Petten, 2008; Polich, 2007) N1, P2 and N400 (partly) reflect linguistic processes (e.g. Dambacher et al., 2006; Duncan et al., 2009; Tarkiainen, Helenius, Hansen, Cornelissen, & Salmelin, 1999). Therefore, only the latter were subjected to further analyses. Peak latency and mean amplitude were determined for both verbs separately. Peaks were semi-automatically determined as the local maximum within 50–200 ms for N1, 100–250 ms for P2 and 300–500 ms for N400. Mean amplitude was computed for the following time windows: N1 (95–135 ms), P2 (160–210 ms) and N400 (300–450 ms). These windows were chosen based on the grand averaged waveforms and previous research (Duncan et al., 2009; Weber-Fox, 2001). To investigate the topographical distribution of the peaks while keeping the amount of data limited, subsets of adjacent electrodes were taken together.

As P2 was observed over the entire scalp, nine clusters with average amplitude/latency of adjacent electrodes were calculated: anterior left (F7, F3), anterior midline (Fz), anterior right (F4, F8), central left (T3, C3), central midline (Cz), central right (C4, T4), posterior left (T5, P3), posterior midline (Pz), and posterior right (P4, T6). Also for the N400, the same nine clusters were created with one exception: posterior left and right did not include T5 and T6 respectively, as no N400 was seen over these electrodes. Since N1 was only seen over posterior regions, one left (T5/P3/O1), one midline (Pz/Oz) and one right (T6/P4/O2) subset was created.

Statistical analyses were performed in IBM SPSS Statistics 19.0. Latency and amplitude were analyzed using repeated measures ANOVAs with three within-subject factors for P2 and N400: hemisphere (left, midline, right), region (anterior, central, posterior) and verbs (action, non-action). As N1 only occurred over posterior sites, the factor 'region' was not included in its analysis. To evaluate whether the assumption of homogeneity of covariance was met, Mauchly's Test of Sphericity was computed for all factors with more than one degree of freedom in the numerator. If the assumption was violated ($\alpha \leq 0.05$), Greenhouse–Geisser (G–G) adjusted *p*-values were used to determine significance. Significance values were set at $\alpha \leq 0.05$ for main and interaction effects. All further pairwise comparisons were Bonferroni corrected.

2.2.5. Source reconstruction

The Statistical Parametric Mapping 8 software package (SPM 8: Welcome Department of Cognitive Neurology, University College, London, United Kingdom) implemented in MATLAB (the MathWorks, Inc., Massachusetts, USA) was used for EEG source reconstruction. To limit the number of comparison, a sensor-space analysis was first performed to search for time points at which maximal differences between action and non-action verbs occurred. In the sensor-space analysis, the ERP data from 0 to 380 ms of every participant was converted into 3D images for every verb category separately. This time window encompasses early and late time points described in similar previous research at which significant sensorimotor activations during action verb processing occurred (Hauk & Pulvermüller, 2004b, Moseley et al., 2013; Pulvermüller et al., 2001). These images were generated by constructing 2D, 64×64 pixels resolution, scalp maps for each time point (using interpolation to estimate the activation between the electrodes) and by stacking the scalp maps over peristimulus time, resulting in [$64 \times 64 \times$ number of time points]-images (Litvak et al., 2011). These images were statistically evaluated by paired *t*-tests. *F*-contrasts were calculated to test for differences of either direction between action and non-action verbs.

The multiple sparse priors (MSP) algorithm (Friston et al., 2008) was used to reconstruct the source activity for every subject and verb category. A 3-layered scalp-skull-brain template head model matched to the MNI template was implemented for which the default electrode positions were used. 8196 Dipoles were assumed on a template cortical surface mesh and the "bemcp" method (BEM) implemented in FieldTrip (Oostenveld, Fries, Maris, & Schoffelen, 2011) was used to calculate the forward model. 3D images containing the evoked energy of the reconstructed activity for every subject and verb class were generated in a time window centred around the significant time point(s) from the sensor-space analysis (Litvak et al., 2011). If the time point occurred before 250 ms post-stimulus onset, a time window of 20 ms was chosen, because processes that take place within the first 250 ms are mostly characterized by short-lived transient activations. If the time point occurred after 250 ms, a window of 40 ms was chosen. Using these images, second level analyses were performed to identify the most significant source areas over subjects and verb class. F-contrasts were calculated by performing paired t-tests for the main effect of verb. As motor related activity was the primary focus of this test, only significant results in frontal and parietal lobe were explored. Because the amount of comparisons was already largely reduced by the sensor-analysis and by limiting the analyses to fronto-parietal areas, *p*-value was set at 0.05 at the source level. The resulting MNI coordinates holding significant activation differences between verb class, were explored by means of the traditional Brodmann categorization and by means of the SPM Anatomy toolbox developed by Eickhoff et al. (2007). Despite the seemingly limited spatial resolution inherent to less dense EEG recordings, low-density recordings have been established to provide an accurate estimate of ERP generators and to be sufficient to fully describe the variance of an ERP data set when compared to high-density recordings (Kayser & Tenke, 2006).

As the pre- and post-central gyrus was the main region of interest, an additional source analysis was performed on the earliest time point with prominent activity above this region. For this purpose, the Global Field Power (GFP) was calculated for each verb class separately over all (pre)frontal and parietal sensors for all participants. GFP illustrates the time course of the overall signal strength of the ERPs. Based on the maps of current estimates that were made for each peak of the GFP, the earliest time point with clear activity over pre- and post-central gyrus was identified. Similar second level analyses were performed on a time window centred around this time point. Again, depending on whether the time point occurred before or after 250 ms post-stimulus onset, a time window of 20 or 40 ms respectively was chosen. *F*-contrasts were calculated by performing paired *t*-tests for the main effect of verb. A mask was applied so only activity in pre- and post-central gyrus (BA 4, 6, 3, 1, 2 and 43) was evaluated. Because only one ROI was included, the criterion for significance could be set at $\alpha = 0.05$.

3. Results

3.1. ERP analysis

The waveforms in Fig. 1 are characterized by a series of components. At posterior electrodes, a very early negative peak around 40 ms was immediately followed by a P1, peaking at around 70 ms, and an N1, peaking at around 115 ms. At anterior sites, the P1 was reversed evoking a negative peak around 70 ms. No equivalent of the posterior N1 was seen. Next, a P2 could be observed over the entire scalp. This wave reached its maximum around 180 ms. The subsequent N2 (peak: 225 ms) and P3 (broad wave around 300 ms were quite small and could only be seen over occipital and T5/T6 electrode sites). Finally, a large N400 occurred, most clearly pronounced over anterior electrodes. Although this component has a protracted morphology, a peak could be described at around 360 ms. Topographic EEG maps for N1, P2 and N400 can be seen in Fig. 2.

Statistical analysis revealed a significant interaction of Verb*Region for the N400 amplitude ($F(2,58) = 3.43$; $\varepsilon = 0.69$; G–G $p = 0.05$). Action verbs showed a larger N400 than non-action verbs over anterior regions ($p = 0.007$). No other significant main or interaction effect of the factor Verb was found for either peak. Furthermore, some distributional amplitude variations were observed. The largest amplitude for the P2 was seen over midline and central electrodes, especially at Cz (Region*Hemisphere: $F(4,116) = 3.28$; $\varepsilon = 0.80$; G–G $p = 0.02$) and for the N400 over left and anterior electrodes(Region*Hemisphere: $F(4,116) = 6.01$; $\varepsilon = 0.67$; G–G $p = 0.001$). The N1 appeared to be smallest (Hemisphere: $F(2,58) = 4.62$; $p = 0.01$) and earliest (Hemisphere: $F(2,58) = 4.13$; $p = 0.02$) over midline electrodes.

3.2. Source reconstruction

Detailed results of the source reconstruction are shown in Table 2. The sensor-space analysis found maximal differences between action and non-action verbs at 228 ms post-stimulus onset

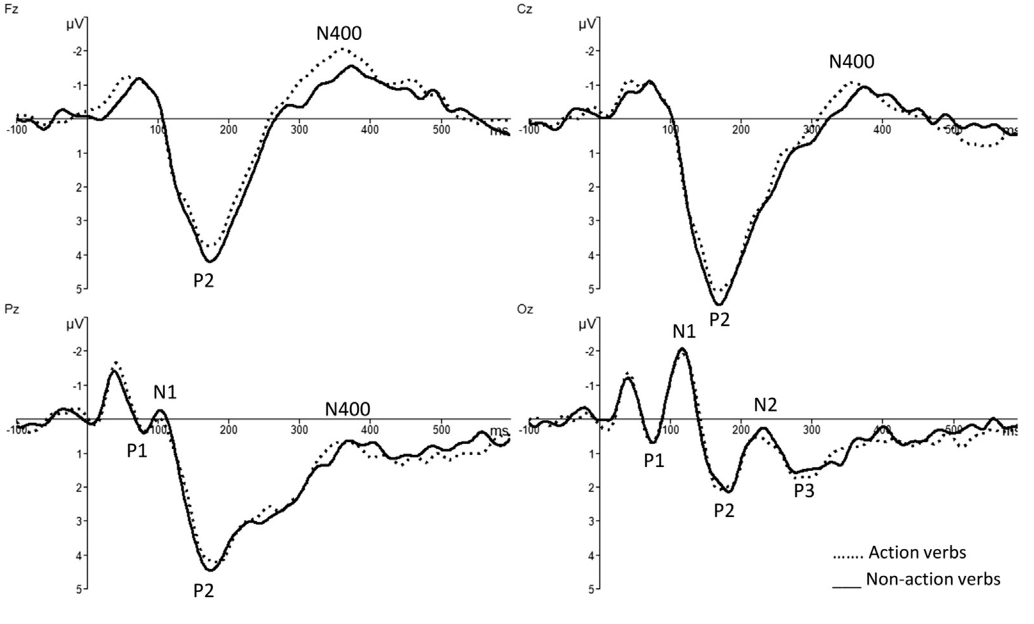

Fig. 1. Grand average for action and non-action verbs separately at midline electrodes. The 100 ms baseline and the first 600 ms of stimulus processing are depicted. Negative is plotted upwards. The *x*-axis represents latency (ms), the *y*-axis represents amplitude (μV).

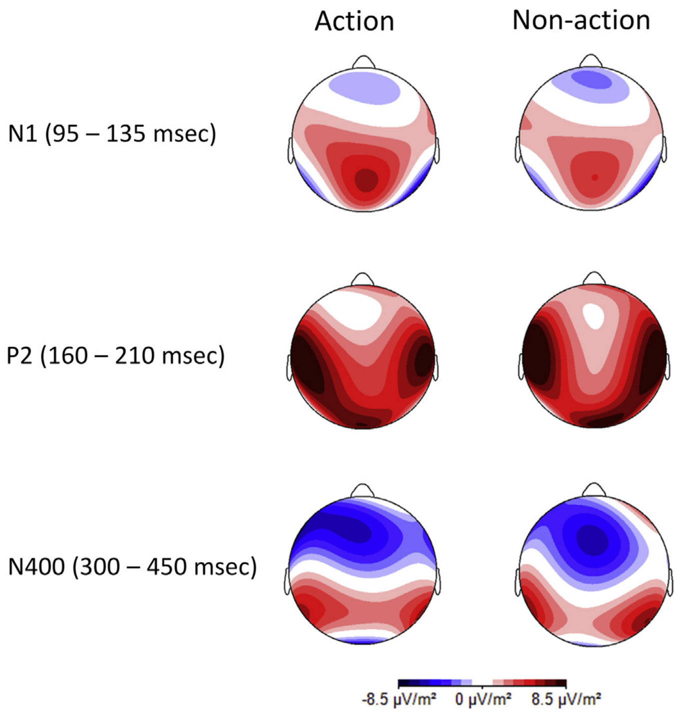

Fig. 2. Topographic EEG maps of N1, P2 and N400.

($F(1,29) = 20.96$, $p < 0.0001$). As this time point occurred within the first 250 ms after stimulus presentation, second level source analyses were performed on an epoch of 20 ms centred around this peak (219–238 ms). Statistical analysis revealed a significant stronger cluster of activation during action compared to non-action verb processing located in dorsolateral prefrontal cortex (DLPFC) in left ($p = 0.011$) and right ($p = 0.011$) hemisphere (see Fig. 3).

Fig. 4 shows the GFP and the map of current estimates for action and non-action verbs separately. The earliest prominent peak activity over pre- and post-central areas occurred at 165 ms for both action and non-action verbs. The subsequent analyses were performed from 155 to 174 ms. During this time window, source reconstruction of the grand averages revealed for both verbs widespread activation in perisylvian regions, including superior temporal cortex and inferior frontal gyrus, supramarginal gyrus, pre- and postcentral gyrus. Also prominent activity was seen in occipital, inferior temporal and

Table 2
Significant results of the source reconstruction for both time windows. Reported are the coordinates of local maxima in MNI space which are part of larger clusters as well as the number of voxels per cluster (2 mm × 2 mm × 2 mm). A description of the region that contains these coordinates (based on both macroscopical parcellation and BA labelling) is added. The last column shows which verb type evoked most energy.

Time interval	Macroscopic anatomical name	BA	Coordinates (MNI)			*p* Value	Extent (voxel)	Highest activation
			x	*y*	*z*			
155–174	R precentral gyrus	BA 6	18	−24	68	0.026	158	Action verb
	L precentral gyrus	BA 4	−14	−28	71	0.027	110	Action verb
219–238	R middle frontal gyrus	BA 9	32	28	42	0.011	164	Action verb
	L middle frontal gyrus	BA 9	−26	26	35	0.011	222	Action verb

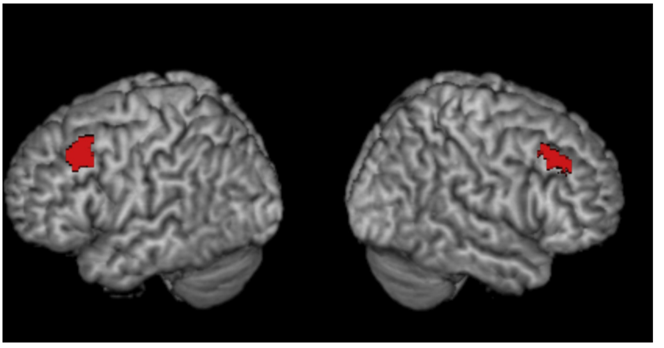

Fig. 3. Result of the sensor-space analysis: source reconstruction for the statistical significant difference between action and non-action verbs in the time window 219–238 ms. An activation focus located in bilateral DLPFC can clearly be identified.

fusiform gyrus (see Fig. 5). Both left and right hemisphere showed similar patterns of activity. Statistical analysis only focused on the sensorimotor cortex (pre- and postcentral gyrus). Indeed, a prominent cluster of activation was located in this region. Action verbs evoked significantly more activation than non-action verbs in left ($p = 0.027$) and right ($p = 0.026$) hemisphere.

4. Discussion

The present study aimed at evaluating the time point(s) of motor related activity in the brain during hand action verb processing. To exclude a possible grammatical class confound, these verbs were contrasted with non-action verbs, i.e. verbs not related to a certain body part or movement. Following previous research, motor related lexico-semantic differences between verbs were not expected in the raw ERP signal (Hauk & Pulvermüller, 2004b; Pulvermüller et al., 2001). Indeed, no statistical difference arose between action and non-action verbs within 300 ms based on the ERP waveforms. Therefore, source reconstruction was performed in two time windows centred around two well-defined time points. Based on the GFP, the earliest, most prominent peak activity over pre- and post-central areas for both verb classes was identified around 165 ms. A sensor-space analysis searched for the time point with maximal difference between action and non-action verbs: 228 ms.

4.1. Motor related activation during action verb processing

An early semantic category effect was observed in bilateral sensorimotor areas. In line with a similar study (Moseley et al., 2013), both action and non-action verb processing evoke the most prominent activation peak at 165 ms post-stimulus onset. Besides a clear activation in sensorimotor cortices, a widespread bilateral activity was observed in core linguistic brain regions, like superior temporal gyrus and inferior frontal gyrus, and in other brain areas partly related to linguistic processing, like fusiform gyrus and supramarginal gyrus. Statistical analyses revealed a significantly higher activation focus in sensorimotor areas during action than during non-action verb processing from 155 to 174 ms. Action verb perception is confirmed to trigger the sensorimotor areas responsible for the motor action the verb refers to. Action verbs and motor action seem to share neuronal representations (Hauk et al., 2004).

The present findings suggest that these neuronal representations contribute essential information to hand action verb processing. The significant sensorimotor difference is observed within 200 ms post-stimulus onset, a time window in which essential lexico-semantic information for concept understanding is retrieved (Federmeier & Kutas, 2001; Hauk et al., 2012; Penolazzi et al., 2007). Moreover, linguistic regions known to be involved in lexical access and semantic retrieval are simultaneously activated (Binder, Desai, Graves, & Conant, 2009). Finally, since both stimulus categories only included

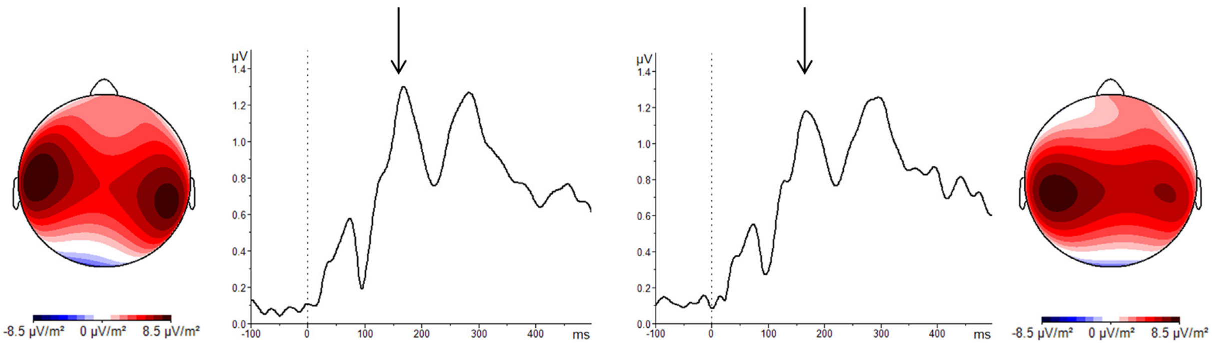

Fig. 4. GFP calculated over all (pre)frontal and parietal sensors for all participants. Non-action verbs are shown on the left, action verbs on the right. The current estimate map at 165 ms post-stimulus onset is presented as well. This is the earliest time point with clear activity over pre- and post-central gyrus.

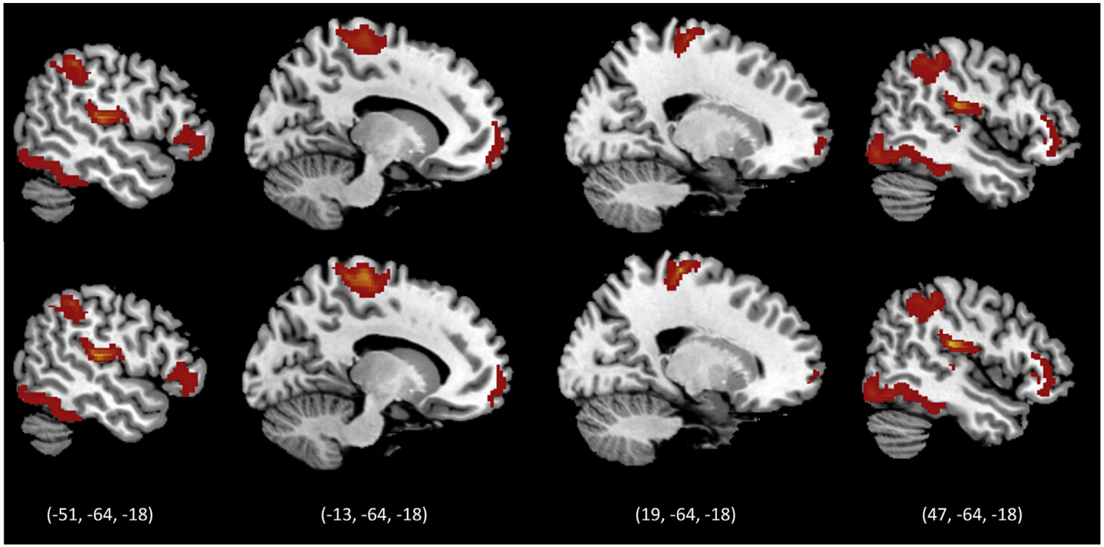

Fig. 5. Source reconstruction of the grand average ERPs from 155 to 174 ms. Four sagittal slices are shown for non-action verbs (top diagram) and action verbs (bottom diagram). The most prominent activation clusters (>2 standard deviations calculated over the whole volume of reconstructed activity) are depicted in red. The two slices on the left are located in the left hemisphere, the two slices on the right are located in the right hemisphere. Corresponding MNI coordinates are shown underneath the pictures (*x*, *y*, *z*). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

verbs, grammatical class will not have confounded the results. The simultaneous activation of sensorimotor regions and other linguistic regions illustrates the functional links that are suggested to exist between the cortical systems for language and action (Pulvermüller, 2005). This is in line with moderate theories of embodied cognition which state that concepts are not only represented in specific language brain areas, but are also modality dependent and grounded in neural action and perception systems (e.g. Barsalou, 1999; Dove, 2009). In sum, these results provide evidence for an early, automatic and functionally relevant role of sensorimotor activation in lexico-semantic processing of hand action verbs.

About 50 ms later, action and non-action verb processing evoked a maximal difference in brain activity. From 219 to 238 ms, bilateral DLPFC was significantly more engaged during action than during non-action verb processing. DLPFC can be seen as a higher order motor region of the brain as it plays an important role in the cognitive control of motor behaviour (e.g. Funahashi, 2001; Miller, 2000; Miller & Cohen, 2001; Tanji & Hoshi, 2001). It receives motor information from both cortical and subcortical motor related brain structures and integrates them for motor control and action planning (Hoshi, 2006). Its involvement in language processing is however not new (Binder et al., 2009; Jeon, Lee, Kim, & Cho, 2009). Moreover, a comparable study found a right prefrontal activation in the same time range. From 210 to 230 ms, hand action verbs evoked higher activations in dorsal DLPFC than leg action verbs (Hauk & Pulvermüller, 2004b). Thus, DLPFC is suggested to contribute to hand action verb processing around this time point. Motor activation during hand action verb processing does not seem to be restricted to the core (pre)motor cortical areas of the brain, but a broad motor brain network is hypothesized to be involved.

Whether DLPFC activation is necessary for action verb understanding cannot be concluded. From 200 to 300 ms onwards, brain activation can be influenced by conscious processes (Dehaene & Changeux, 2011). Since DLPFC activation occurred on the border of this time range, the present data do not allow a straightforward conclusion. Future studies with the present task in motor pathologies might address this issue.

4.2. *Concreteness/imageability effects*

The present action verbs are significantly more imaginable than the non-action verbs. This argument is often used to posit that motor related activations during action verb processing are rather related to mental imagery and concreteness effects (Postle et al., 2013; Tomasino et al., 2008). Although especially concreteness cannot entirely be excluded to have influenced the present results, several arguments are in favour for the embodied cognition point of view.

Explicit mental imagery can be excluded because this also involves posterior brain regions which were not found to be more engaged during action verb processing (Willems, Toni, Hagoort, & Casananto, 2009). Furthermore, effects of mental imagery are reported to occur from 300 (Gullick, Mitra, & Coch, 2013), 500 (West & Holcomb, 2000), or even 700 ms (Welcome, Paivio, McRae, & Joanisse, 2011) onwards.

A ‘concreteness effect’ is however present in the N400: action verbs evoke a significantly larger N400 than non-action verbs over anterior brain regions which is compatible with concrete words evoking larger N400 amplitudes than abstract words (e.g. Gullick et al., 2013). The N400 is generally accepted to reflect semantic processing, more specifically it represents the integration of different kinds of information in large scale networks (Hauk et al., 2012; Kutas & Federmeier, 2000). The larger N400 for action verbs would be related to a larger amount of neural correlates that need to be integrated (Xiao, Zhao, Zhang, & Guo, 2012). Notwithstanding a concreteness effect can be seen, it does not seem responsible for the difference in activation. Post-lexical processing is suggested to be necessary to elicit concreteness effects (West & Holcomb, 2000). Indeed, concreteness is typically found to modulate ERP results from 250 to 300 ms onwards (Barber, Otten, Kousta, & Vigliocco, 2013; Kanske & Kotz, 2007; Palazova, Manthill, Sommer, & Schacht, 2011), just as in the current study. By contrast, sensorimotor and DLPFC activity occurred earlier, at 165 and 228 ms respectively. Moreover, from 250 ms onwards, no significant difference between both verb classes appeared in the source reconstruction. Thus, no significant difference in brain activation was found in a time window where a concreteness effect is (1) generally found in previous research, and (2) also found in the present N400 results. Consequently, a concreteness influence is very unlikely to occur in preceding time windows (Kanske & Kotz, 2007; West & Holcomb, 2000). Moreover, the verbs used in the Hauk and Pulvermüller (2004b) study, that observed a comparable result, did not significantly differ in concreteness/imageability. Thus even with similar concreteness scores, hand action verbs elicited higher DLPFC activity than other verbs.

4.3. Bilateral motor related activation

The significantly higher activation during action verb processing was observed in left and right hemisphere for both DLPFC and sensorimotor cortex. The laterality of activation during action word processing has been suggested to be determined by language dominance and/or handedness (Hauk & Pulvermüller, 2011; Willems, Hagoort, & Casasanto, 2010). Although only right handed participants were included, both uni- and bimanual words were presented which may be responsible for the observed bilateral activity. In general, hand action verbs are more often reported to activate left and right motor areas than action verbs related to other body parts (Hauk & Pulvermüller, 2004b; Pulvermüller et al., 2001; Raposo et al., 2009; Rüschemeyer, Brass, & Friederici, 2007; Tettamanti et al., 2005).

4.4. Sensorimotor involvement in non-action verb processing

The source reconstruction of the grand average of the non-action verbs also showed an important sensorimotor activity from 155 to 174 ms. Thus, even abstract verbs not related to a certain body part or body movement evoke some activity in sensorimotor cortex. Considering the early time point of activation and the simultaneous activation of brain regions involved in lexical access and semantic retrieval, this activity seems to be related to lexico-semantic processing as well.

Sensorimotor contribution in lexico-semantic processing of abstract verbs is subject for discussion. While some theories claim that only linguistic brain areas contribute to abstract verb processing, others propose a reliance on modal (perception and action related) information as well (e.g. Borghi & Cimatti, 2010; Fodor, 1998; Louwerse & Jeuniaux, 2010; Paivio, 1986). Most neuroimaging studies report activations in language related areas like inferior frontal and middle temporal gyrus (Wang, Conder, Blitzer, & Shinkareva, 2010). However, some studies also found limited sensorimotor activity (Rodriguez-Ferreiro, Gennari, Davies, & Cuetos, 2010; Sakreida et al., 2013). In a recent fMRI study, strong haemodynamic responses to abstract emotion words were observed in face- and arm-related motor regions. The authors concluded that these abstract words evoked precentral activity because they refer to body internal states. As face and arms are used to express emotions, these brain regions are activated when the corresponding words are perceived (Moseley, Carota, Hauk, Mohr, & Pulvermüller, 2012). Thus, depending on the semantic associations of a word, sensorimotor brain regions can be activated during abstract word processing. The present non-action verbs were primarily used as contrast condition and were not controlled for strict semantic features. They were a conglomeration of words, not characterized by one strong semantic association. Therefore, the diverse semantic networks across non-action items might have resulted in some sensorimotor engagement that is however weaker than the one evoked by the action verbs (Moseley et al., 2013).

Other researchers link the sensorimotor activity to the acquisition modality of abstract words. When a child obtains language, the meaning of an abstract word has to be explained linguistically. This would be a bodily experience as well which might lead to face-related motor activity during abstract word processing (Borghi, Flumini, Cimatti, Marocco, & Scorolli, 2011; Scorolli et al., 2012). This research group also reports arm-related sensorimotor activity during abstract word processing in a later post-lexical stage (Scorolli et al., 2012).

4.5. Early ERP results

As mentioned above, no statistical differences arose between action and non-action verbs within 300 ms based on the ERP waveforms. Latency and amplitude of N1 and P2 were similar for both verb classes. Besides reflecting early sensory processes, both peaks show important amplitude modulations by lexical and psycholinguistic features. As N1 is related to visual word form processes (Brem et al., 2006; Salmelin, 2007; Tarkiainen et al., 1999), its amplitude is affected by psycholinguistic features like word length and neighbourhood size (Hauk & Pulvermüller, 2004a; Hauk, Davis, et al., 2006; Hauk, Patterson, et al., 2006). The exact function of the P2 remains unresolved. However, recent studies link it with early lexico-semantic and -syntactic processes as its amplitude is modulated by word frequency and grammatical class (Dambacher et al., 2006; Palazova et al., 2011; Takashima et al., 2001). Since stimuli were carefully controlled for these confounds and both stimuli types were verbs, no statistical difference in amplitude was expected.

4.6. Additional considerations

The strength of EEG research concerns its excellent temporal resolution. The major aim of this study was to clarify the time points at which significant differences can be found between action and non-action verbs. To interpret these timing results, one should have a look at their spatial characteristics as well. For this purpose, the source reconstruction was performed. Spatial resolution of EEG research is however limited and therefore, its results should be interpreted with caution. To meet this limited spatial resolution, no fine grained analyses nor interpretations were performed (e.g. making a distinction between ventral and dorsal DLPFC) though this might have provided valuable information (Hoshi, 2006).

Two concerns might arise regarding the participants: (1) large age range, and (2) unequal number of men and women. Because age is suggested to influence information processing rate (Cerella, 1985; Salthouse, 1991) and language differences between men and women have (inconsistently) been described (Gölgeli et al., 1999; Swink & Stuart, 2012; Wirth et al., 2007), additional statistical analyses were performed to evaluate a possible impact of age and gender on the present results. For age, all analyses were performed again without the oldest 6 participants (older than 1 SD above the mean age of the entire group). As these results mirrored the original results entirely, the large age range can be concluded to not have influenced the present findings.

A possible gender effect was explored by using a linear mixed model approach as this technique is preferred if data are not balanced (Field, 2009). Although men and women showed slight differences in the ERP analyses, these differences were rather small as there were no significant results when men and women were compared to each other. For the source reconstruction, both genders evoked similar results in the areas of interest. These findings do not support the suggestion that the results of the original group are biased by the results of the men.

5. Conclusion

Single verb processing evoked the most prominent peak activity at 165 ms after word presentation. Besides a clear activation in several linguistic brain regions, also bilateral sensorimotor cortex was engaged. This sensorimotor activation was significantly higher during action than during non-action verb processing from 155 to 174 ms. This result is suggested to be a word-specific semantic difference as (1) a grammatical class confound can be excluded (2) it occurs within 200 ms in which essential lexico-semantic information is known to be retrieved, and (3) brain regions involved in lexical access and semantic retrieval are simultaneously activated. From 219 to 238 ms, action compared to non-action verbs evoked significantly higher DLPFC activations which is a higher order motor brain region involved in action planning.

Hand action verbs thus seem to activate the motor programmes of the actions the verbs refer to. This is not restricted to the core (pre)motor cortical areas of the brain. A broad motor brain network is hypothesized to be involved. While the sensorimotor activation appears to be automatic and necessary for action verb understanding, this cannot be concluded for DLPFC activation. Nevertheless, the present results are in line with embodied theories of cognition which state that concepts are represented in specific language brain areas and in neural action and perception systems. Future EEG research in disorders affecting the motor system may contribute to our understanding of motor related brain activations during hand action verb processing.

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Appendix A

List of all stimuli items: hand- and arm related action verbs on the left, non-action verbs on the right. Although several words are polysemous in English, they are not in Dutch.

Action verbs		Non-action verbs	
aaien	to stroke	bedriegen	to cheat
aangeven	to hand	behoren	to belong
aanraken	to touch	beloven	to promise
boetseren	to mould	blijken	to turn out
boksen	to box	dempen	to muffle
borduren	to embroider	dulden	to tolerate
borstelen	to brush	dunken	to deem
breien	to knit	durven	to dare
duwen	to push	eisen	to demand
gieten	to pour	ergeren	to annoy
gooien	to throw	flitsen	to flash
grijpen	to grab	geloven	to believe
haken	to crochet	genieten	to enjoy
kammen	to comb	gokken	to gamble
klappen	to clap	gunnen	to grant
kloppen	to knock	haten	to hate
kneden	to knead	hopen	to hope
knijpen	to pinch	huren	to rent
knippen	to cut (with a scissor)	kiezen	to choose
krabben	to scratch	kwellen	to agonize
masseren	to massage	lenen	to lend
naaien	to sew	liegen	to lie, as in tell a lie
pakken	to grasp	melden	to report
plukken	to pick	menen	to mean
roeren	to stir	missen	to miss
schetsen	to sketch	mogen	to may
schilderen	to paint	onthouden	to remember
schillen	to peel	opletten	to pay attention
schrijven	to write	pleiten	to plead
schrobben	to scrub	raden	to guess
schuren	to sand	rijmen	to rhyme
slaan	to hit	riskeren	to risk
smeren	to butter	roesten	to rust
smijten	to fling	schamen	to be ashamed
snijden	to cut (with a knife)	schatten	to estimate
stempelen	to stamp	scheiden	to separate
strelen	to caress	schijnen	to shine
strijken	to iron	spijten	to regret
tekenen	to draw	stralen	to shine
tikken	to tap	treiteren	to torment
timmeren	to hammer	treuren	to grieve
trekken	to pull	twijfelen	to doubt
typen	to type	verlossen	to release
vangen	to catch	verstaan	to understand
vasthouden	to hold	verwachten	to expect
werpen	to cast	verzinnen	to make up
wijzen	to point	verzoeken	to request
wrijven	to rub	vrezen	to fear
wuiven	to wave	wachten	to wait
zwaaien	to wave	wensen	to wish

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